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MSI MAG Z690 TOMAHAWK WIFI Motherboard

Great board for getting on the DDR5 bandwagon

The latest addition to MSI's TOMAHAWK series is the MSI MAG Z690 TOMAHAWK WIFI. The Tomahawk line of boards has historically provided a great balance between features and price sitting right between MSI's premium and budget lineup of motherboards. With the Intel 12th Gen Alder Lake processors, the boards have gotten a lot more expensive thanks to the addition of PCIe Gen 5.0 and the inclusion of DDR5 memory support. Tomahawk boards have been spotted for as low as 9K and the highest we've seen till now was 22-24K but with the 12th Gen boards, we have the MSI MAG Z690 TOMAHAWK WIFI retailing for nearly 30K. You can still get the DDR4 variant for 25K but these boards are way more expensive than they used to be. However, given the features that they pack, is it really too much?

BUILD QUALITY

The MSI MAG Z690 TOMAHAWK WIFI comes with a pretty standard layout. At the top, there are the VRM heatsinks which are huge chunks of aluminium and are split into two halves. There is no heat pipe running through the heatsinks. For the VRM circuitry, the board uses an MP2960 by Monolithic Power Systems. From the power stages around the processor socket, it seems to be in a 16+1+1 configuration with 16 phases going into the CPU, 1 phase for the iGPU and 1 phase for AUX. Thus far, MSI has been using a Duet Rail configuration with their Z690 boards and this one is no different. Each of the power stages happens to be Monolithic Power Systems MP87992 which supports up to 70 A of current. Then there's the CPU socket area which has expanded with the Intel 12th Gen processors. It's now 78x78 mm in area rather than the 75x75 mm



configuration that we'd seen with the previous generation. This means that the CPU cooler mount holes have spread out further outwards from the CPU socket. This isn't a problem but we see the holes are a little too close to the heatsinks.

Speaking of heatsinks, the board is littered with them since the additional PCIe lanes allow for a lot more NVMe SSDs to be accommodated on 12th Gen Intel motherboards. MSI added a new snap-lock mechanism for mounting NVMe SSDs. You simply need to insert the SSD and then twist the lock to hold the outer edge of the SSD rather than screwing it in. It's more convenient now.

Beside the NVMe SSDs lies the PCH heatsink which is quite massive and is made of aluminium but the right edge of the board has been cut away around the PCH heatsink by a little less than a centimetre. Also, some of the SATA HDD data connectors have been pushed under the PCH heatsink. Attached and detaching SATA cables to these connectors isn't an issue so we'll look past this one.

Lastly, the board uses the Realtek

ALC4080 audio CODEC, has plenty of USB 3.2 ports and features 2.5G Ethernet along with Wi-Fi 6E.

PERFORMANCE

The MSI BIOS hasn't changed with the new 12th Gen platform aside from including entries for the new DDR5 memory configuration and the added NVMe SSDs. You have all the options needed to play around with the memory configuration including switching Gears, altering the individual timings or just letting the BIOS figure out the nitty-gritty of configuring your memory. Comparing the memory bandwidth with a DDR5 kit set to 4800 MT/s and 5200 MT/s, we found little difference in performance from some of the other flagship boards we've tested.



PERFORMANCE.....78
FEATURES.....81
BUILD QUALITY.....74
VALUE FOR MONEY...75

VERDICT

The MSI MAG Z690 TOMAHAWK WIFI is certainly the most expensive TOMAHAWK board we've tested from MSI but considering the

sheer amount of connectivity options available thanks to the new Intel 12th Gen platform, it is certainly worth the money. Considering that most Z690 DDR5 boards start around the same price point, the MSI MAG Z690 TOMAHAWK WIFI is a great board to build your 12th gen system on. If the price seems a tad high, you can always check out the B660 boards which have nearly the same set of features as the Z690 ones.

—Mithun Mohandas

SPECIFICATIONS

PLATFORM: Intel Z690 | MEMORY SUPPORT: 128GB DDR5 | MEMORY SLOTS: 4 | PCIe X16 SLOTS: 3 | LAN: Intel i225-V 2.5G | WI-FI: Intel Wi-Fi 6E | AUDIO: Realtek ALC4080 | WARRANTY: 3 years.

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